ROCSCIENCE

Case4
Slide2 - An Interactive Slope Stability Program
Date Created: 13/03/2026, 8:34:42 am
Software Version: 9.041

Table of Contents

Project Summary 3
General Settings 4
Analysis Options 5
Groundwater Analysis 6
Random Numbers 7
Surface Options 8
Seismic Loading 9
Materials 10
Global Minimums 11
Method: bishop simplified 11
Method: spencer 11
Valid and Invalid Surfaces 12
Method: bishop simplified 12
Method: spencer 12
Error Code Descriptions 12
Slice Data 13
Global Minimum Query (bishop simplified) - Safety Factor: 1.23469 13
Global Minimum Query (spencer) - Safety Factor: 1.23141 14
Interslice Data 15
Global Minimum Query (bishop simplified) - Safety Factor: 1.23469 15
Global Minimum Query (spencer) - Safety Factor: 1.23141 16
Entity Information 17
Group 1 17
Shared Entities 17

Slide2 Analysis Information

Case4

Project Summary

File Name:	Case4.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:18.975s
Project Title:	Slide2 - An Interactive Slope Stability Program
Date Created:	13/03/2026, 8:34:42 am

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Maximum
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	25
Tolerance:	0.0001
Maximum number of iterations:	100
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	30
Circles per division:	15
Number of iterations:	15
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Materials

Material 1	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	16 kN/m3
Cohesion	9 kPa
Phi	32 °
Water Surface	None
Ru Value	0

Global Minimums

Method: bishop simplified

FS	1.234690
Center:	21.048, 67.275
Radius:	43.212
Left Slip Surface Endpoint:	30.001, 25.000
Right Slip Surface Endpoint:	58.077, 45.000
Resisting Moment:	55233.7 kN-m
Driving Moment:	44734.9 kN-m
Total Slice Area:	113.732 m2
Surface Horizontal Width:	28.0761 m
Surface Average Height:	4.05087 m

Method: spencer

FS	1.231410
Center:	22.581, 64.313
Radius:	39.995
Left Slip Surface Endpoint:	30.020, 25.016
Right Slip Surface Endpoint:	57.604, 45.000
Resisting Moment:	50783.9 kN-m
Driving Moment:	41240.5 kN-m
Resisting Horizontal Force:	1024.72 kN
Driving Horizontal Force:	832.156 kN
Total Slice Area:	113.181 m2
Surface Horizontal Width:	27.5847 m
Surface Average Height:	4.10304 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	32802
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	29693
Number of Invalid Surfaces:	3109

Error Codes

Error Code -108 reported for 308 surfaces
Error Code -111 reported for 2801 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).
-111 = Safety factor equation did not converge

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.23469

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	1.12304	5.79431	12.72	Material 1	9	32	8.8854	10.9707	3.15381	0	3.15381	5.15947	5.15947
2	1.12304	17.0977	14.2514	Material 1	9	32	13.2864	16.4046	11.8498	0	11.8498	15.2245	15.2245
3	1.12304	27.8247	15.7934	Material 1	9	32	17.3454	21.4162	19.8701	0	19.8701	24.7762	24.7762
4	1.12304	37.9628	17.3472	Material 1	9	32	21.0667	26.0108	27.223	0	27.223	33.8036	33.8036
5	1.12304	47.4974	18.9143	Material 1	9	32	24.4532	30.1921	33.9144	0	33.9144	42.2935	42.2935
6	1.12304	56.4121	20.4963	Material 1	9	32	27.5072	33.9629	39.949	0	39.949	50.2315	50.2315
7	1.12304	64.6881	22.0947	Material 1	9	32	30.2299	37.3246	45.3289	0	45.3289	57.6008	57.6008
8	1.12304	72.3043	23.7115	Material 1	9	32	32.6217	40.2777	50.0548	0	50.0548	64.3825	64.3825
9	1.12304	79.2366	25.3486	Material 1	9	32	34.6817	42.8212	54.1253	0	54.1253	70.5553	70.5553
10	1.12304	85.4575	27.0082	Material 1	9	32	36.4084	44.9531	57.537	0	57.537	76.0946	76.0946
11	1.12304	90.936	28.6927	Material 1	9	32	37.799	46.6701	60.2848	0	60.2848	80.9729	80.9729
12	1.12304	95.6366	30.4048	Material 1	9	32	38.8498	47.9675	62.3611	0	62.3611	85.1585	85.1585
13	1.12304	99.5185	32.1474	Material 1	9	32	39.5558	48.8392	63.7561	0	63.7561	88.6151	88.6151
14	1.12304	102.535	33.9241	Material 1	9	32	39.9109	49.2776	64.4577	0	64.4577	91.3011	91.3011
15	1.12304	104.632	35.7388	Material 1	9	32	39.9074	49.2733	64.4508	0	64.4508	93.1682	93.1682
16	1.12304	105.746	37.5958	Material 1	9	32	39.5364	48.8152	63.7177	0	63.7177	94.1602	94.1602
17	1.12304	105.803	39.5004	Material 1	9	32	38.7871	47.89	62.237	0	62.237	94.2111	94.2111
18	1.12304	104.715	41.4589	Material 1	9	32	37.6467	46.482	59.9837	0	59.9837	93.2426	93.2426
19	1.12304	102.378	43.4785	Material 1	9	32	36.1005	44.5729	56.9285	0	56.9285	91.1609	91.1609
20	1.12304	98.6616	45.5682	Material 1	9	32	34.1311	42.1413	53.0372	0	53.0372	87.8521	87.8521
21	1.12304	93.4096	47.739	Material 1	9	32	31.7183	39.1623	48.2698	0	48.2698	83.1755	83.1755
22	1.12304	86.4231	50.0047	Material 1	9	32	28.8387	35.6069	42.58	0	42.58	76.9544	76.9544
23	1.12304	73.0319	52.383	Material 1	9	32	24.2645	29.9591	33.5416	0	33.5416	65.0304	65.0304
24	1.12304	46.1305	54.8978	Material 1	9	32	16.3239	20.155	17.8517	0	17.8517	41.0764	41.0764
25	1.12304	15.8877	57.5816	Material 1	9	32	8.04102	9.92817	1.48538	0	1.48538	14.147	14.147

Global Minimum Query (spencer) - Safety Factor: 1.23141

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	1.10339	5.80576	11.525	Material 1	9	32	12.0387	14.8246	9.32127	0	9.32127	11.776	11.776
2	1.10339	17.129	13.1433	Material 1	9	32	17.278	21.2763	19.6462	0	19.6462	23.6807	23.6807
3	1.10339	27.87	14.7723	Material 1	9	32	21.8653	26.9251	28.6861	0	28.6861	34.4518	34.4518
4	1.10339	38.0161	16.4136	Material 1	9	32	25.85	31.8319	36.5387	0	36.5387	44.1534	44.1534
5	1.10339	47.553	18.069	Material 1	9	32	29.2751	36.0496	43.2884	0	43.2884	52.8395	52.8395
6	1.10339	56.4639	19.7401	Material 1	9	32	32.1777	39.6239	49.0084	0	49.0084	60.5551	60.5551
7	1.10339	64.7298	21.4288	Material 1	9	32	34.5899	42.5944	53.7623	0	53.7623	67.338	67.338
8	1.10339	72.3289	23.1374	Material 1	9	32	36.5399	44.9956	57.605	0	57.605	73.2188	73.2188
9	1.10339	79.2361	24.8681	Material 1	9	32	38.052	46.8576	60.5848	0	60.5848	78.2222	78.2222
10	1.10339	85.423	26.6233	Material 1	9	32	39.1472	48.2063	62.7432	0	62.7432	82.3666	82.3666
11	1.10339	90.8567	28.406	Material 1	9	32	39.8441	49.0644	64.1164	0	64.1164	85.6654	85.6654
12	1.10339	95.4996	30.2193	Material 1	9	32	40.1584	49.4515	64.736	0	64.736	88.1268	88.1268
13	1.10339	99.3083	32.0666	Material 1	9	32	40.1042	49.3847	64.6291	0	64.6291	89.7538	89.7538
14	1.10339	102.232	33.9521	Material 1	9	32	39.6933	48.8787	63.8193	0	63.8193	90.5445	90.5445
15	1.10339	104.213	35.8804	Material 1	9	32	38.9359	47.946	62.3266	0	62.3266	90.4912	90.4912
16	1.10339	105.181	37.857	Material 1	9	32	37.8407	46.5974	60.1685	0	60.1685	89.5811	89.5811
17	1.10339	105.054	39.8881	Material 1	9	32	36.4154	44.8423	57.3597	0	57.3597	87.7949	87.7949
18	1.10339	103.733	41.9815	Material 1	9	32	34.6664	42.6886	53.9131	0	53.9131	85.1067	85.1067
19	1.10339	101.099	44.1463	Material 1	9	32	32.5994	40.1432	49.8396	0	49.8396	81.4817	81.4817
20	1.10339	97.0032	46.3937	Material 1	9	32	30.2193	37.2123	45.1491	0	45.1491	76.8755	76.8755
21	1.10339	91.2598	48.7382	Material 1	9	32	27.5307	33.9016	39.8508	0	39.8508	71.2306	71.2306
22	1.10339	83.629	51.1981	Material 1	9	32	24.5385	30.2169	33.9541	0	33.9541	64.4717	64.4717
23	1.10339	72.7824	53.7977	Material 1	9	32	21.0402	25.9091	27.0602	0	27.0602	55.8055	55.8055
24	1.10339	47.9112	56.5707	Material 1	9	32	15.1002	18.5946	15.3545	0	15.3545	38.2297	38.2297
25	1.10339	16.5783	59.5659	Material 1	9	32	8.68245	10.6917	2.70723	0	2.70723	17.486	17.486

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.23469

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	30.0005	25.0004	0	0	0
2	31.1236	25.2539	9.1792	0	0
3	32.2466	25.5392	20.7203	0	0
4	33.3697	25.8568	33.8882	0	0
5	34.4927	26.2076	47.997	0	0
6	35.6158	26.5925	62.4081	0	0
7	36.7388	27.0123	76.5291	0	0
8	37.8618	27.4682	89.8132	0	0
9	38.9849	27.9614	101.759	0	0
10	40.1079	28.4934	111.912	0	0
11	41.231	29.0659	119.865	0	0
12	42.354	29.6805	125.26	0	0
13	43.4771	30.3395	127.794	0	0
14	44.6001	31.0453	127.219	0	0
15	45.7231	31.8006	123.353	0	0
16	46.8462	32.6088	116.086	0	0
17	47.9692	33.4735	105.388	0	0
18	49.0923	34.3993	91.33	0	0
19	50.2153	35.3914	74.096	0	0
20	51.3383	36.4564	54.0135	0	0
21	52.4614	37.6019	31.5879	0	0
22	53.5844	38.8378	7.5524	0	0
23	54.7075	40.1764	-17.0585	0	0
24	55.8305	41.6338	-38.6923	0	0
25	56.9536	43.2316	-48.8832	0	0
26	58.0766	45	0	0	0

Global Minimum Query (spencer) - Safety Factor: 1.23141

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	30.0195	25.0156	0	0	0
2	31.1229	25.2406	11.1854	7.19153	32.7385
3	32.2263	25.4982	25.1868	16.1936	32.7386
4	33.3297	25.7892	40.9647	26.3379	32.7386
5	34.4331	26.1142	57.6092	37.0394	32.7387
6	35.5364	26.4742	74.326	47.7873	32.7387
7	36.6398	26.8702	90.4238	58.1372	32.7386
8	37.7432	27.3032	105.306	67.7053	32.7385
9	38.8466	27.7747	118.461	76.1634	32.7386
10	39.95	28.2861	129.459	83.2349	32.7388
11	41.0534	28.8392	137.948	88.6927	32.7387
12	42.1568	29.436	143.648	92.357	32.7386
13	43.2602	30.0786	146.351	94.0948	32.7386
14	44.3635	30.7699	145.923	93.8199	32.7386
15	45.4669	31.5128	142.306	91.4943	32.7386
16	46.5703	32.311	135.519	87.1307	32.7386
17	47.6737	33.1686	125.667	80.7963	32.7386
18	48.7771	34.0908	112.948	72.6191	32.7387
19	49.8805	35.0836	97.6688	62.7953	32.7386
20	50.9839	36.1546	80.2589	51.6018	32.7387
21	52.0872	37.313	61.2989	39.4116	32.7386
22	53.1906	38.5707	41.5556	26.7178	32.7386
23	54.294	39.9429	22.0361	14.1679	32.7386
24	55.3974	41.4504	4.45798	2.86622	32.7386
25	56.5008	43.1219	-4.54692	-2.92341	32.7387
26	57.6042	45	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	95, 20 95, 45 55, 45 30, 25 10, 25 10, 20